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ABSTRACT

Video Object Segmentation (VOS), an important part of the understanding of videos, is vital for multiple industries, including surveillance and security systems, self-driving cars, human-computer interface design, and augmented reality. The main aim of VOS is to detect and separate moving objects in video frames adequately to extract necessary details from the subsequent processing steps. But the problem of how to provide constant segmentation and detection of each class in real conditions has not been solved. Situations with a complex lighting, with changes within several frames, with shadows, reflections, occlusions, or rapidly changing backgrounds make it difficult to segment an object and frequently results in errors and inconsistency. This project, "Invisio: "Video Object Segmentation in Complex Environment, deals with these challenges using machine learning and computer vision techniques. The subsequent and more complex deep learning models allow for effective operation with video frames and include features such as attention maps, optical flows, and global motion aggregation. These augmentations improve how the model reasons about a case and reinforces dependable segmentation of different videos. Specifically, outdoor videos often suffer from constrained visibility due to different environmental conditions and other small particles in the air that scatter the light in the atmosphere.[10] Combining the two problems of Video Object Segmentation and Tracking can overcome their respective difficulties and improve their performance. Based on the results achieved in this project, the state-of-art for segmentation is seen to have been substantially improved. Enhanced performance under adverse conditions affirms the need for applying this system in general uses. The findings of this work serve as a starting point to further improvements of video object segmentation procedures that will unlock many integrations of video analysis, security and privacy.